

WEEK 5 PROGRESS REPORT

WHO DEFINES THE PROBLEM? A COMPUTATIONAL ANALYSIS OF HOW EU INSTITUTIONS FRAME ARTIFICIAL INTELLIGENCE REGULATION

27 September 2026

Project GitHub Link: <https://github.com/michelle-qyb/SOSC314-Course-Project/>

Division of Responsibilities

For this project stage, each group member was responsible for the following components:

Tim Maurice Steiner:

Primary responsibility: Diagnostic design and validity assessment — term-influence testing, bootstrap resampling, disentangled sensitivity analysis, written deliverables.

Specific tasks completed this week: Built the diagnostics module and driver (diagnostics.py, 05_diagnostics.py); ran the four checks reported below; revised the dictionary and the Week 4 interpretation in light of them; wrote the progress report and figure.

Yuhan Qiu:

Primary responsibility: Topic-model diagnostics — stability, specification sensitivity and uncertainty of the LDA and NMF models, and their reconciliation with the dictionary results.

Specific tasks completed this week: Built 05_topic_diagnostics.ipynb (preprocessing comparison, six-seed LDA stability, $K = 20$ sensitivity, paragraph- versus document-weighting, document bootstrap); wrote the topic-model results and Figure 2.

Why These Diagnostics

Week 4 reported that the frame dictionary was stable under six representation specifications while Fightin' Words lost a fifth to a third of its top terms, and treated this as a reason to trust the dictionary more. That comparison was incomplete in a way worth correcting, because it made the dictionary look robust along the one axis it cannot move. The six specifications all varied preprocessing — which words are removed before counting — and the dictionary counts only terms on its own seven lists, so words it ignores cannot disturb it. Three of the six (tf-idf, bigrams, minimum document frequency) live inside a vectoriser the dictionary never builds. Its stability was therefore close to a restatement of what it does not look at. This week tests the axis that can move it, the content of the lists, together with two questions Week 4 did not ask: whether the corpus is large enough to support the differences we read off it, and which preprocessing decision was actually doing the work. The topic models from Week 4 are put through the same checks, together with two that only they need: stability across random starts and across the number of topics.

1. Term Influence

Refitting the profile with each of the 152 dictionary terms removed in turn shows that the measure is not an aggregate of many weak signals. Removing digital alone shifts a frame share by 6.6 percentage points; data by 5.8, market by 4.9, risk by 3.9 and rights by 3.7. Forty-one terms move some share by more than a point. The terms with most influence are also the least frame-specific: digital, data, model and development are the common vocabulary of EU technology policy rather than markers of an innovation frame, and market appears in any internal-market text.

Removing eleven such terms from their frames shifts shares by up to 15.4 points and, unlike every preprocessing variant tested in Week 4, breaks the rank order of frames. Technological innovation falls from

35.4 to 20.0 per cent for the Commission and from 27.2 to 15.6 for the Council. The Week 4 observation that technological innovation dominates every institution was therefore substantially an artefact of four generic words, and the report's instruction not to over-interpret that bar was better founded than we knew. The topic models show the same thing from the other side: the LDA topic closest to technological innovation is led by *digital*, *development*, *technologies*, *data* and *innovation*, and the Commission's lead on it is not robust (25.0 per cent, interval 20.7 to 29.3, against Parliament's 21.7, 16.6 to 27.3). Regulatory governance rises correspondingly, and the Council's lead on it widens from 20.0 against 11.0 to 27.5 against 15.6.

2. Term Overlap

Week 4 stated that the frames are not independent because their term lists overlap, citing jobs and risk. Checking the lists directly, three pairs share terms and no pair shares more than one: jobs across economic opportunity and labour, autonomy across fundamental rights and security, training across labour and technological innovation. Risk appears only in consumer protection. The Week 4 claim was therefore wrong on its example and overstated in general, and it is corrected here. Non-independence is a real property of framing — a paragraph can carry several frames — but it does not arise from the dictionary construction to any material degree.

3. Sampling and Model Stability

The corpus is the population of retrievable AI-related documents rather than a sample from one, so a confidence interval is not a sampling-error estimate in the survey sense. Resampling documents with replacement within each institution answers a narrower and more useful question: how much of an observed difference depends on which particular documents an institution happened to produce? Figure 1a reports 95 per cent intervals from a thousand resamples.

Only one difference survives cleanly. The Commission leads security (8.0, 6.0 to 10.3) above both others, whose intervals reach only 5.6 and 4.7. Its lead on technological innovation (35.4, interval 31.9 to 38.9) clears the Council's interval (27.2, 22.9 to 31.2) but not Parliament's (24.5 to 33.9). The Council's lead on regulatory governance, the clearest institutional result in Week 4, carries an interval from 12.6 to 27.9 that overlaps Parliament's and almost touches the Commission's; on twenty-eight documents it cannot be stated as a finding. Parliament's lead on fundamental rights (10.3 to 19.5) overlaps the Council's (7.8 to 20.1) substantially. The Council's labour and social share, 10.7 with an interval from 3.2 to 18.9, carries no information at all.

The topic models were put through the same resampling and, for LDA, six further random seeds and a refit at $K = 20$. The rights topic recurs in none of the six refits and its lead passes to the Council at $K = 20$. The one topic-model result that survives every check is the Council's lead on the LDA topic covering high-risk systems and their requirements (19.9 per cent, 11.6 to 29.1, against Parliament's 5.8, 2.8 to 9.8; held in all six refits and at $K = 20$). We read this as a difference in content rather than framing. The NMF topic that does match governance vocabulary (conformity assessment, notified bodies) is small and not robust. The Council spends more of its text on the substance of the high-risk rules; that does not show it frames AI as a governance problem.

4. Which Preprocessing Decision Mattered

Week 4 measured all six specifications against the baseline, so the three that vary counting also carried the stopword change and appeared to cost as much as it did. Measuring each against its own reference separates them. Removing the procedural and rhetorical stopword lists replaces 30 per cent of the top-20 terms. Adding bigrams to the cleaned text replaces 17. Switching to tf-idf replaces none at all, and raising the minimum

document frequency from five to twenty replaces one per cent. The Week 4 remark that a purely technical choice such as tf-idf moves the answer as much as the deliberate stopword decisions was an artefact of the confounded comparison and does not hold. Harmonising the topic models onto the same stopwords lists moved LDA considerably (four of ten topics keep a counterpart, against eight of ten for NMF). Switching their shares from paragraph- to document-weighting, as the dictionary uses, more than halves the Council's share of the high-risk topic, from 41.6 to 19.9 per cent.

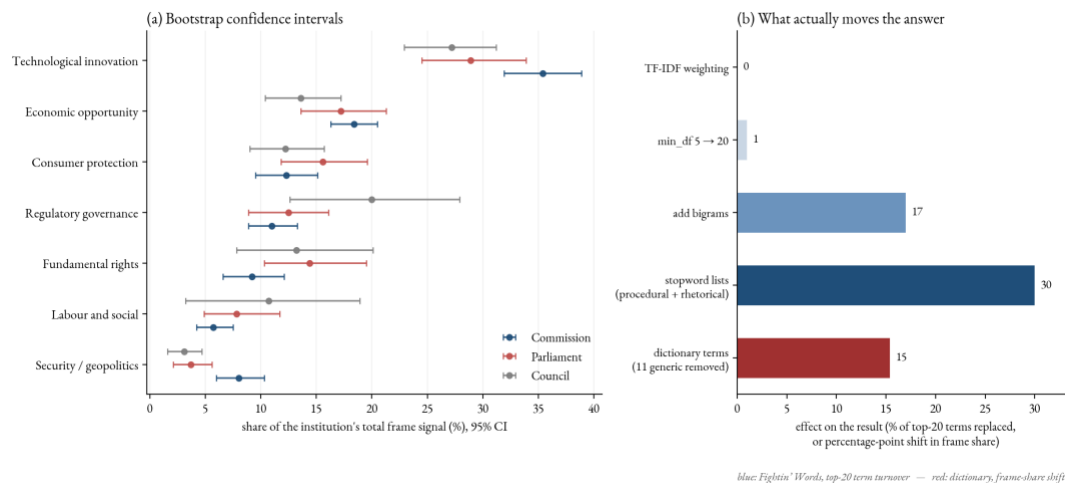


Figure 1. Panel (a): frame profile with 95% bootstrap confidence intervals from 1,000 resamples of documents within institution. Panel (b): the effect of each analytic decision measured against its own reference — top-20 term turnover for *Fightin' Words* (blue), percentage-point shift in frame share for the dictionary (red).

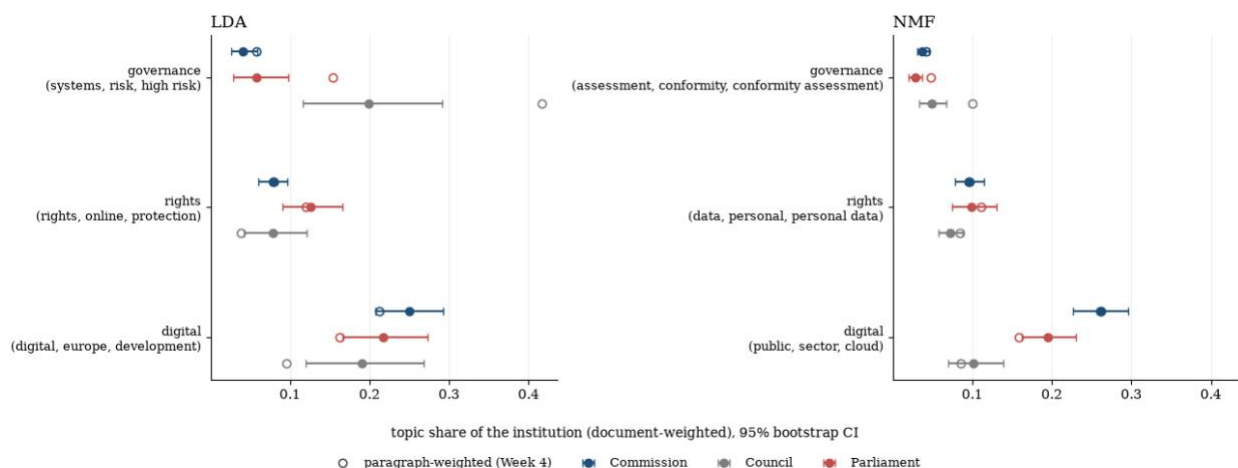


Figure 2. Tracked topics in LDA (left) and NMF (right): document-weighted shares with 95% bootstrap intervals from 1,000 resamples of documents within institution; hollow markers show the Week 4 paragraph-weighted values. NMF has no clean rights topic at $K = 10$, so data protection stands in.

Revisions

Several changes follow. The Council governance result is demoted from a finding to a pattern awaiting more documents, as is Parliament's rights lead, which the topic models also fail to reproduce. Applied consistently, the non-overlap criterion also rules out the Commission's technology lead, whose interval overlaps Parliament's, so security remains the only robust dictionary lead. The topic models add one robust result, the Council's attention to the high-risk regime, which we report as a difference in content rather than in framing. The dominance of technological innovation is withdrawn as a substantive observation and reported as a

property of four generic terms. The tf-idf claim is withdrawn. The tightened dictionary is committed alongside the reported one rather than replacing it, because choosing the specification that produces the cleaner result after seeing both is precisely the practice these diagnostics exist to catch; which is preferable is a question for validation, not for us.

Limitations and Next Steps

None of this is validation against ground truth. The diagnostics establish which words the measure depends on, that its frames are near-independent, that some differences exceed what the corpus supports, and which processing decision matters — but not that a paragraph scoring high on a frame reads that way to a human. That remains the gate, and the leave-one-term-out results now tell us where to aim the hand-coding: paragraphs scoring high on technological innovation, where the generic-term problem is concentrated, and paragraphs scoring high on regulatory governance, where transparency and accountability may be picking up legal boilerplate rather than a governance frame. The same sample should include Council paragraphs high on the LDA high-risk topic, to test whether they read as governance framing or as statutory text.